Lab Experiment 3

Ethical Use of Generative AI In Engineering Decision Making

Learning Outcomes

By the end of these activities, students will be able to:

1. Explain the role of Generative AI in assisting engineering decision making processes.
2. Identify lecture terms found in AI tools such as bias, hallucinations, and overreliance.
3. Differentiate between appropriate and inappropriate uses of AI in academic and professional engineering contexts.
4. Apply AI tools to revise or support original engineering work without violating academic integrity guidelines.
5. Reflect on the personal responsibility of engineers when integrating AI into their professional practice.

Lab Exercise

Objective:

To explore the ethical use of Generative AI in engineering decision making by applying AI tools responsibly, evaluating their outputs, and reflecting on the importance of accuracy, transparency, and integrity in professional practice

Instructions:

Complete the following activities referring to chapter 8 and other appropriate resources in your lab report paper:

Lab activity overview:

1. Select a research article

- Visit CUNY Academic Works – Publications & Research collection

  CUNY Academic Works, Publications and Research | City University of New York (CUNY)

- Select a published research article of your liking (Preferably AI related) For example:

PDF "ChatGPT Told Me to Say It": AI Chatbots and Class Participation Apprehension in University Students, Daisuke Akiba

PDF Arduino-ESP32 based Smart Irrigation System, Ahmed D. Hassebo, Kevin B. Montes, and Erick Cabrera

PDF Hybridize Functions: A Tool for Automatically Refactoring Imperative Deep Learning Programs to Graph Execution, Raffi Khatchadourian, Tatiana Castro Vélez, Mehdi Bagherzadeh, Nan Jia, and Anita Raja

2. Understanding the article

- Read and understand the selected research articles as best as you can.
- Summarize in your own words the methodology and key findings of the article.
- Use a Generative AI tool to revise your summary by uploading or referencing the PDF of the article in the prompt.
- Compare the two versions, what differences do you notice between your original summary and the revised version? Did the AI introduce any errors, omissions or changes in emphasis?

3. Identify AI opportunities

Once you familiarize yourself with the article, answer the following questions:

- Could generative AI assist in interpreting or improving aspects of this research? If so, how?

For example: Simplifying complex language, generating code templates, proposing alternative approaches.

- Following on the previous question: what are potential risks, like bias, hallucinations or overreliance that you could encounter when using in that context?

Reflection and transparency

Prepare a brief reflection that includes:

- The AI tool used and how
- Your evaluation: benefits vs. ethical concerns.
- Suggestions for responsibly integrating AI into engineering workflows like this.

Grading Rubric

- **Objectives:** Identifies objectives and provides the necessary background or context (0-3 points)
- **Results/Analysis:** Presents results clearly. Analyzes findings correctly, with proper calculations and/or comparisons as needed. (0-6 points)
- **Discussion & Reflection:** Explains significance of results, discusses errors and connects outcomes to engineering/technology concepts. (0-5 points)
- **Professionalism & Format:** Report is well-organized, follows the provided template, uses clear language, proper grammar and formatting. Includes citation and references (if outside sources were used) (0-6 points)
- **Total:** 20 points

Deliverables

- Submit your completed activity responses as instructed by your instructor.